

Hanyi Zhao

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Research Interests

My research focuses on robotic visual perception and planning in dynamic, unstructured environments, aiming to enable robots to understand, adapt, and manipulate effectively across diverse real-world scenarios.

Education

• Harbin Institute of Technology, Shenzhen

Sep 2022 - present

- Degree: B.E. in Automation; GPA: 91.758/100
- Awards: National Scholarship, First-Class Academic Scholarship (two times)

Publications

- [1] **H. Zhao***, J. Zhu*, Z. Yan*, Y. Li, Y. Deng[†], and X. Wang[†], “Learning generalizable language-conditioned cloth manipulation from long demonstrations”, *International Conference on Intelligent Robots and Systems (IROS)*, 2025. [\[Paper\]](#) [\[Website\]](#) [\[Code\]](#)

Experience

• Center for Artificial Intelligence and Robotics, Tsinghua SIGS

Oct 2023 - Feb 2025

Research Intern with [Prof. Xueqian Wang](#)

Shenzhen, China

- Developed a novel pipeline that extracts reusable skills from existing long demonstrations and composes them to generalize across unseen multi-step cloth manipulation tasks.
- Introduced an autonomous skill discovery framework for cloth manipulation, leveraging an LLM to provide commonsense knowledge for task decomposition.
- Conducted extensive experiments in both simulation and real-world settings, validating the effectiveness and generalization of the proposed approach.

• Adaptive Computing Lab & Smart Systems Institute, NUS

Feb 2025 - Present

Research Intern with [Prof. David Hsu](#)

Singapore

- Implemented a real-time 3D scene graph framework that fuses object meshes, poses, and inter-object relations through integrated detection, segmentation, tracking, pose estimation, and 3D reconstruction.
- Introduced a tracking module that fuses end-effector motion with observations to accurately estimate and update object poses during manipulation.
- Utilized spatial relationships within the scene graph to preserve the consistency and continuity of moving objects through temporary invisibility or occlusion.

Projects

• RoboMaster Aerial Robot Design and Optimization

Nov 2022 - Aug 2023

RoboMaster Competition 2023

- Designed and built a lightweight, high-performance UAV featuring a shooting gimbal, an airframe, propeller guards, and other custom structural components.
- Constructed the full system using advanced prototyping techniques such as 3D printing, laser cutting, and precision machining.

• Autonomous Mobile Robot Design and Development

Dec 2023 - Jul 2024

The 19th National University Students Intelligent Car Race

- Designed and implemented an autonomous mobile robot, capable of efficiently performing diverse tasks such as trajectory following, target detection, and visual classification.
- Developed the complete mechanical, control, and perception system, including motion control, serial communication, and image processing algorithms, enabling adaptive and accurate task execution in complex environments.

Skills

- **Programming Languages:** Python, C++, C, Matlab
- **Softwares & Tools:** Linux, ROS, PyTorch, OpenCV, Solidworks, Latex, Windows
- **Hardware:** Multiple Sensors, OptiTrack Motion Capture System, STM32, Basic Mechanical Design and Assembly
- **Languages:** English (fluent: IELTS 7.5), Chinese (native)

Awards & Certifications

- H Award of the Mathematical Contest in Modeling (MCM)

May 2024

- First Prize in Guangdong Province of the Blue Bridge Cup National Software and Information Technology Professional Talent Competition

Jun 2024

- Second Prize in the South China Division of the 19th National University Students Intelligent Car Race

Jul 2024